

AutoPartFlow ERP

Automobile Spare Parts Distribution & Sales Management System

GROUP NUMBER
IS 24

DEGREE PROGRAMME
B.Sc. in Information Systems, UCSC

TEAM SIZE
4 Members

1. Executive Summary

The initial proposal defense identified that an all-encompassing, multi-domain ERP spanning unrelated retail sectors was overly broad, prone to superficial implementation, and carried high scope risk for a four-member team within a one-year timeframe. In response, the project underwent a deliberate strategic pivot: rather than building a generic system, the scope has been re-anchored onto a single, high-friction vertical — **Automobile Spare Parts Distribution and Sales Management (AutoPartFlow ERP)**.

The system reflects real operational workflows for regional spare-part distributors serving both business-to-business (B2B) workshops and walk-in retail (B2C) clients. By discarding non-essential ERP overhead — recruitment, complex HR, and multi-tier logistics — the project concentrates effort on three high-value domain features: an intelligent vehicle-to-part compatibility lookup engine, a mobile-optimized point-of-sale (POS) interface for field representatives, and real-time inventory synchronization across all sales and order channels.

2. Panel Feedback Synthesis & Evaluation Analysis

Overly Broad & Undefined Scope

A generalized ERP meant to serve grocery, pharmacy, and hardware sectors diluted requirements; without a specialized domain, modules would only be built superficially.

Reinventing Standard Wheels

Generic inventory forms and invoice layouts already exist in abundant open-source software — custom development lacked a distinctive domain problem to justify it.

Feasibility & Bandwidth

Extensive HR, recruitment workflows, and wide-ranging admin capabilities exceeded the realistic capacity of a four-person undergraduate group alongside coursework.

Ambiguity of the "Smart" Label

Calling the platform "Smart" required functional justification beyond conventional CRUD data-entry screens and generic database storage.

Absence of Real-World Client Realism

Without a defined operational client, the risk was building disjointed features that did not reflect actual industry sales and supply workflows.

3. Strategic Pivot: Action Taken

I. Single Specialized Domain

- Re-anchored entirely on automobile spare-part distribution networks.
- Scoped for a mid-scale distributor: central warehouse, field sales reps, and counter sales.
- Cut recruitment, appraisals, payment gateways, GPS tracking, and multi-warehouse logistics.

II. Concrete Domain Innovation

- **Vehicle-to-Part Compatibility Engine** — a multi-tier mapping matrix (Brand → Model → Year → Engine/Fuel type) that eliminates incorrect part ordering.
- **Mobile-First Field Sales POS** — real-time stock checks and instant quote/invoice generation on-site at repair shops.
- **Dual-Channel Ordering** — links back-office, field reps, and a self-service customer portal.

III. Realistic Workload Balance

- Workload split into four distinct, vertical business roles.
- Each member owns a full module: UI, business logic, and backend persistence.
- Clear entity ownership avoids overlapping development.

4. Workload Distribution

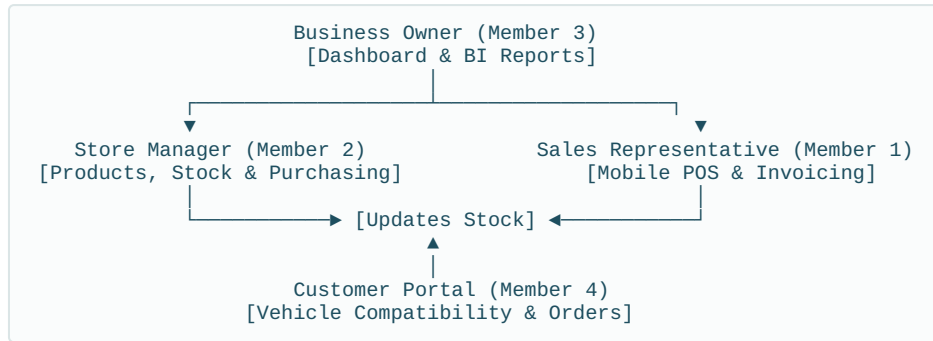


Figure 1 — Functional ownership and stock-update data flow across the four role modules.

MEMBER	ROLE	CORE RESPONSIBILITIES	MODULES & SCREENS	DATABASE ENTITIES
Member 1	Sales Representative	• Mobile-first POS for on-site visits • B2B & B2C customer profile management • Cart, instant invoicing, payment recording • Order status transitions & sales logs	Sales Rep Dashboard, Mobile POS, Customer Management, Order Processing, Invoicing, Sales History	customers, sales, sale_items, orders, order_items, payments
Member 2	Store Manager	• Spare part records, categories, brands • Vehicle compatibility logic engine • Stock movements & low-stock alerts • Supplier profiles & purchase orders	Product Management, Inventory & Stock Adjustments, Purchase Orders, Supplier Management, Compatibility Engine	products, categories, brands, vehicles, compatibility, inventory, stock_movements, suppliers, purchase_orders, purchase_order_items
Member 3	Business Owner / Admin	• Session management, hashing, RBAC • Employee records & system settings • Executive dashboard: revenue, inventory value, rep performance • Consolidated audit & activity reports	Authentication & RBAC, Owner Dashboard, Employee Accounts, BI Reports, Activity Logs, System Settings	users, roles, employees, notifications, activity_logs, settings
Member 4	Shop Customer / Public User	• Public catalog with make/model/year filtering • Spare Part Finder with live stock & pricing • Digital order submission for garages • Delivery status tracking & fulfillment	Public Product Catalog, Vehicle-Based Part Finder, Customer Ordering Portal, Order Tracking, Delivery Management	product_catalog (view), compatibility (integration), customer_orders, deliveries